

Write-Heavy System Capacity Example

Write-Heavy System Example (IoT Ingestion)

Assumptions:

- DAU (devices) = 10M sensors
- Events per device per day = 10,000
- Event size = 200 bytes

Step 1. RPS:

- Writes: $(10M \times 10,000) / 86,400 \approx 1.15M$ RPS

Step 2. Throughput:

- $1.15M \times 200B \approx 230$ MB/sec

Step 3. Daily Storage:

- $230 \text{ MB/sec} \times 86,400 \approx 19.9$ TB/day

Step 4. Peak Load:

- Writes $\approx 3.5M$ RPS (3× avg)

Key Observation:

- Extreme write pressure, storage and streaming backbone critical.